

Data Flow Diagram

FitBuddy – AI Fitness Plan Generator using Gemini Models

Level-1 Data Flow

User → index.html → FastAPI /generate-workout → Gemini Pro → 7-day workout plan.

FastAPI → Gemini Flash → nutrition/recovery tip.

FastAPI → SQLite/SQLAlchemy → stores user and plan data.

User feedback → /submit-feedback → Gemini Pro → revised plan → SQLite → result.html.

Admin request → /view-all-users → SQLite → all_users.html.

Data Entities

User: user ID, name, age, weight, goal, intensity.

Plan: original generated plan and updated plan.

AI outputs: workout plan and nutrition/recovery tip.